Course One
Foundations of Data Science

Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- ☐ Complete the PACE Strategy Document to plan your project while considering your audience members, teammates, key milestones, and overall project goal.
- ☐ Create a project proposal for the data team.

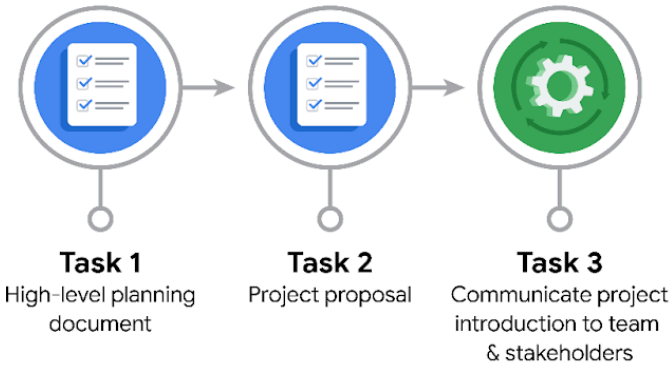
Relevant Interview Questions

Completing this end-of-course project will empower you to respond to the following interview topics:

- As a new member of a data analytics team, what steps could you take to get 'up to speed' with a current project? What steps would you take? Who would you like to meet with?
- How would you plan an analytics project?
- What steps would you take to translate a business question to an analytical solution?
- Why is actively managing data an important part of a data analytics team's responsibilities?
- What are some considerations you might need to be mindful of when reporting results?

Reference Guide

This project has three tasks; the following visual identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- Who is your audience for this project?

TLC team members and stakeholders

- What are you trying to solve or accomplish? And, what do you anticipate the impact of this work will be on the larger needs of the client?

To make regression model that helps estimate taxi fares before the ride,that will help users to know how much the fare before the ride

- What questions need to be asked or answered?

Is there will be changes in the financial state in the country ?

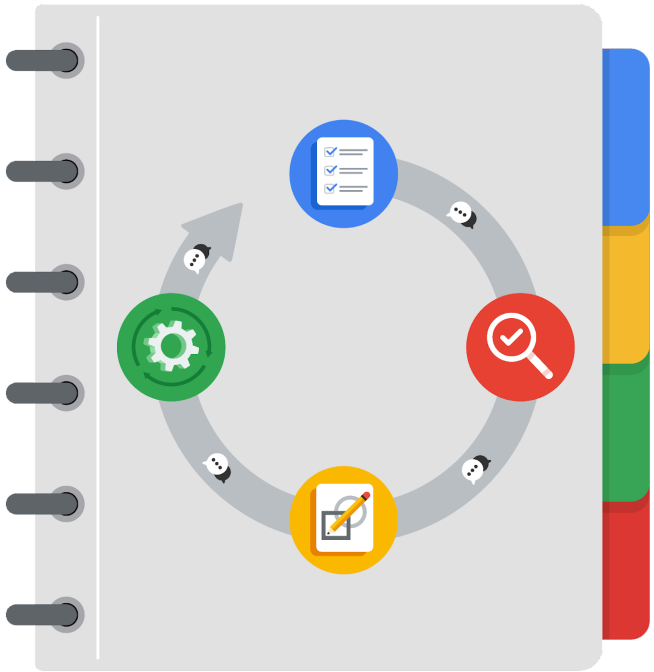
- What resources are required to complete this project?

The trips data for the past years that include the distant and the price of the trip

- What are the deliverables that will need to be created over the course of this project?

A model the get an expected price for the ride befor it start

THE PACE WORKFLOW



[Alt-text: The PACE Workflow with the four stages in a circle: plan, analyze, construct, and execute.]

You have been asked to demonstrate for the company's data team how you would use the PACE workflow to organize and classify tasks for the upcoming project. Select a PACE stage from the dropdown buttons. A few tasks involve more than one stage of the PACE workflow. Additionally, not every workplace scenario will require every task. Refer back to the Course 1 end-of-course portfolio project overview reading if you need more information about the tasks within the project.

Project tasks

Following are a group of tasks your company's data team has determined need to be completed within this project. The data analysis manager has asked you to organize these tasks in preparation for the project proposal document. First, identify which stage of the PACE workflow each task would best fit under using the drop down menu. Next, give an explanation of why you selected the stage for each task. Review the following readings to help guide your selections and explanation: The PACE stages and Communicate objectives with a project proposal. You will later reorder these tasks within a project proposal.

1. **Evaluating the model:** Execute

   Why did you select this stage for this task?

   After the model has been constructed , the data run through to see if the model is meet's the project expectations

2. **Conduct hypothesis testing:** Construct **and** Analyze

   Why did you select these stages for this task?

   During the analyze stage we put the statistics that will used and in construct the test execute

3. **Begin exploring the data:** Analyze

   Why did you select this stage for this task?

   In analyze stage we start to collect the data and explore it

4. **Data exploration and cleaning:** Analyze **and** Plan

   Why did you select these stages for this task?

   In plan phase we determine the methods to explore the data , then we start to cleaning it in analyze phase

5. **Establish structure for project workflow (PACE):** Plan

Why did you select this stage for this task?

The First thing in the project is to determine the workflow

6. **Communicate final insights with stakeholders:** Execute

Why did you select this stage for this task?

After the construction of the model we start to communicate with the stakeholders as the final stage

7. **Compute descriptive statistics:** Analyze

Why did you select this stage for this task?

The compute of the statistics take place in the analyze stage

8. **Visualization building:** Construct **and** Analyze

Why did you select these stages for this task?

IN analyze stage we start to build visualization forme the data after preparing it and in the construct phase we make visualization form the model

9. **Write a project proposal:** Plan

Why did you select this stage for this task?

After determining the plan and how the project will go we write the project proposal to know what steps will take through the project

10. Build a regression model: Construct and Analyze

Why did you select this stage for this task?

After preparing the data we start to make the regression model and what the parameters will it take in the analyze phase, then we start to build the model in the construction phase

11. Compile summary information about the data: Analyze

Why did you select this stage for this task?

In analyze stage we make the summary after clean it

12. Build machine learning model: Construct

Why did you select this stage for this task?

In construct phase we start to build the model